Arno Strouwen

Statistical Consultant

Curriculum Vitae
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Profile

Statistician specialized in experimental design for dynamic systems and adaptive Bayesian design. Consulting through Strouwen Statistics on research-driven projects in industry and academia.

Experience

2022–Present **Founder and Principal Consultant**, Strouwen Statistics

- Consultancy in experimental design, statistical modelling, and scientific computing in Julia, focused on optimal data gathering strategies.

2025–Present **Senior Product Engineer**, PumasAI, Contract through Strouwen Statistics

- Development of software for noncompartmental analysis (NCA), bioequivalence, in vitro–in vivo correlation (IVIVC), and PBPK and PBBM modelling.
- **NCA**: validated against Phoenix WinNonlin, the industry reference, on **248 concentration profiles** covering **74 NCA parameters**.
- Contributing to the JuliaStats open-source organization.

2022–Present **Lecturer**, KU Leuven, Substitute for Prof. Peter Goos

- Teaching the course **Experimental Design** in the Master of Statistics and Data Science, including a new lecture on **Bayesian optimization**.

2022–2025 **Research Scientist**, JuliaHub, Contract through Strouwen Statistics

- Successfully onboarded several leading pharmaceutical companies to the PumasQSP software.
- Significant contributor to the SciML documentation and to **SciMLSensitivity**, which computes derivatives of differential equation solutions with respect to their parameters.

2021–2022 **Postdoc Statistician**, The Janssen Pharmaceutical Companies of Johnson & Johnson

- **Model Based Design of Experiment**: designing **10x accelerated stability studies** to precisely predict the **shelf life** of pharmaceutical drugs and vaccines.
- **Bayesian Optimization**: high-throughput optimization of chemical reactions, supervising a PhD intern.
- **Probabilistic Programming**: Bayesian Non-linear **mixed effect modelling** of powder flowability.

2016–2021 **Ph.D. Fellow Strategic Basic Research**, Research Foundation Flanders

- **Optimal Experimental Design for Dynamic Systems:** Developing novel robust experimental design methodology for dynamic systems with both measurement and process noise.
- **Metabolism of Pear During Hypoxia**: Designing informative experiments to study respiration and fermentation characteristics of pear fruit.

Skills

DOE **Model-Based Design**, Optimal Design, Screening Designs, Response Surface Designs, Mixture Experiments, Blocked and Split-Plot Experiments, Bayesian Experimental Design, Adaptive Design, Bayesian Optimization

ML **Deep Learning**, Reinforcement Learning, Normalizing Flows, Diffusion, Variational Bayes, Amortized Inference, Gaussian Processes, Probabilistic and Differentiable Programming

Statistics **Bayesian Statistics**, Linear and Nonlinear Mixed Effects Models, Small Sample Inference, Bayesian and Kalman Filtering, Uncertainty Quantification, Information Theory

SciML **Scientific Computing**, Differential Equations, Sensitivity Analysis, Numerical Optimization, Learning Missing Physics, Universal Differential Equations, Symbolic Regression

Pharma **Pharmacometrics**, NCA, Bioequivalence, IVIVC, PBPK, PBBM, Stability Studies, Reaction Engineering

Programming **Julia** (Enzyme.jl, Reactant.jl, Lux.jl, SciML, JuliaStats), JMP; *basic knowledge: Python, SQL, R, Matlab*

Peer Reviewed Publications

2026 **Deep Adaptive Bayesian Screening.**
Jade Lejeune Herman, Arno Strouwen, Johan A. K. Suykens and Peter Goos, *Transactions on Machine Learning Research*, accepted, arXiv 2607.16927.

2026 **Deep Adaptive Model-Based Design of Experiments.**
Arno Strouwen and Sebastian Micluţa-Câmpeanu, *CDC 2026*, to appear, arXiv 2603.16146.

2026 **Bayesian Symbolic Regression for Missing Physics.**
Arno Strouwen, *IFAC World Congress 2026*, arXiv 2603.14918.

2025 **Experimental Design for Missing Physics.**
Arno Strouwen and Sebastian Micluţa-Câmpeanu, *DYCOPS 2025*.

2023 **Adaptive and Robust Experimental Design for Linear Dynamical Models using Kalman filter.**
Arno Strouwen, Bart Nicolaï and Peter Goos, *Statistical Papers*, 64 (4).

2022 **Robust Dynamic Experiments for the Precise Estimation of Respiration and Fermentation Parameters of Fruit and Vegetables.**
Arno Strouwen, Bart Nicolaï and Peter Goos, *PLOS Computational Biology*, 18 (1).

2021 **D- and I-optimal design of multi-factor industrial experiments with ordinal outcomes.**
Karel Van Brantegem, Arno Strouwen and Peter Goos, *Chemometrics and Intelligent Laboratory Systems*, 221.

2019 **A Note on the Output of a Coordinate-Exchange Algorithm for Optimal Experimental Design**
Arno Strouwen and Peter Goos, *Chemometrics and Intelligent Laboratory Systems*, 192.

2019 **Optimizing Oxygen Input Profiles for Efficient Estimation of Michaelis-Menten Respiration Models.**
Arno Strouwen, Bart Nicolaï and Peter Goos, *Food and Bioprocess Technology*, 12 (5), 769-780.

Selected Presentations

2026 **Deep Adaptive Design of Experiments for Dynamical Systems.**
JuliaCon 2026 and CDC 2026

2026 **InVitroInVivoCorrelation.jl: A Modular Julia Package for Level A IVIVC.**
PAGE 2026 (poster) and NCS 2026

2026 **Bayesian Optimization as Design of Experiments.**
ENBIS-26

2026 **From Optimal Designs to Optimal Policies.**
Tutorial, mODa 14

2025 **Small Sample Inference for Mixed Models.**
ENBIS-25, JuliaCon 2025

2025 **Experimental Design for Missing Physics.**
ENBIS-25, DYCOPS 2025, JuliaCon 2025 and Leuven Statistics Days 2025

2024 **Bayesian non-linear mixed effects model for safer powder storage.**
JuliaCon 2024, ENBIS-24, and NCS 2024

2023 **Optimal Design for Model Autocompletion.**
ENBIS-23

2023 **Adaptive and Robust Experimental Design for Linear Dynamical Models using Kalman filter.**
mODa 13

2022 **Model Based Experimental Design for Accelerated Small Molecule Stability Studies**
NCS 2022

2018 **Optimizing an Oxygen Input profile to Estimate Michaelis-Menten Respiration Parameters**
ENBIS Spring Meeting 2018

2017 **Optimal Design of Experiments for Non-Linear Models using JMP**
KU Leuven Seminar

Education

2022–2023 **Selected graduate coursework in Statistics and Data Science**, UHasselt, Belgium

2016–2021 **Ph.D. in Bio-science engineering**, KU Leuven, Belgium

- **Thesis:** "*Optimal Design of Dynamic Experiments in Bioscience Engineering*" under supervision of **Prof. Peter Goos** and **Prof. Bart Nicolaï**

2011–2016 **B.Sc.&M.Sc. in Bioscience-engineering**, *Bio-systems engineering*, magna cum laude, KU Leuven, Belgium